Aarhus University

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Parallax - Game Design

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1. Outline

The following is an outline of the story and overall gameflow of **Parallax**.

1.1. Story

In **Parallax**, players take on the roles of a man and his cat as they navigate everyday environments from two very different perspectives. The game's asymmetric gameplay is based on the natural differences between how humans and cats perceive and interact with the world. Objects that appear important to the human may simply look like toys to the cat, while dangers, pathways, or hiding spots may only stand out to one of them.

The story begins inside their home, where the kitchen and living room act as introductory levels that establish the characters' unique abilities. The man can move furniture, operate objects, and interact with the environment directly, while the cat can climb, crawl through tight spaces, and reach places inaccessible to humans.

As the pair venture outside, ordinary locations become playgrounds for cooperation and exploration. A construction site introduces moving machinery, unstable structures, and vertical traversal challenges, while a playground emphasizes climbing, agility, and navigation through oversized obstacles. Eventually, the journey leads to a tall office building, where increasingly complex environments force both players to rely on communication and teamwork to progress.

At its core, the game focuses on cooperation between two characters who experience the same world differently, encouraging players to communicate clearly and use each character's strengths to overcome obstacles together.

1.2. Game flow

The players control the human and the cat respectively, while they navigate through obstacle courses throughout everyday environments. The goal of each level is the navigate through the level, with the help of each other, to reach the exit zone.

The players might find boxes that only the human can move, or keys that only the cat can reach, and as such, must cooperate to get both players from the start of the level, to the end.

For each completed level, the player is awarded a trophy for the role they completed the level as, and a possible elusive dev-trophy, should they beat the best time of the development team. Their time is registered, and displayed on the level selection scene, to incentivize players to push for better times.

2. Characters

The following section describes the characters of **Parallax** and their abilities

2.1. Backstory - Cat

The cat, shown in **Figure 1** is a curious indoor pet living together with its owner in a small apartment. Much of the world is viewed through instinct and playfulness, causing ordinary objects to appear exciting, unfamiliar, or worth investigating. The cat naturally explores spaces the human cannot reach and reacts strongly to sounds, movement, and danger.

2.1.1. Relation to gameplay

The cat's perspective emphasizes exploration, traversal, and observation. Small spaces, elevated areas, and hidden pathways are accessible to the cat, encouraging players to communicate information the human cannot easily discover alone. Many objects also appear differently to the cat, reinforcing the asymmetric gameplay experience.

2.1.2. Controls

The cat is controlled using fast and agile movement focused on jumping, climbing, crawling through vents, and interacting with smaller objects. The controls are designed to feel lightweight and responsive to reflect feline movement.

2.1.3. Abilities

- Climb furniture and vertical surfaces
- Crawl through tight spaces and vents
- Reach elevated platforms
- Better visibility in dark areas
- Faster movement and traversal

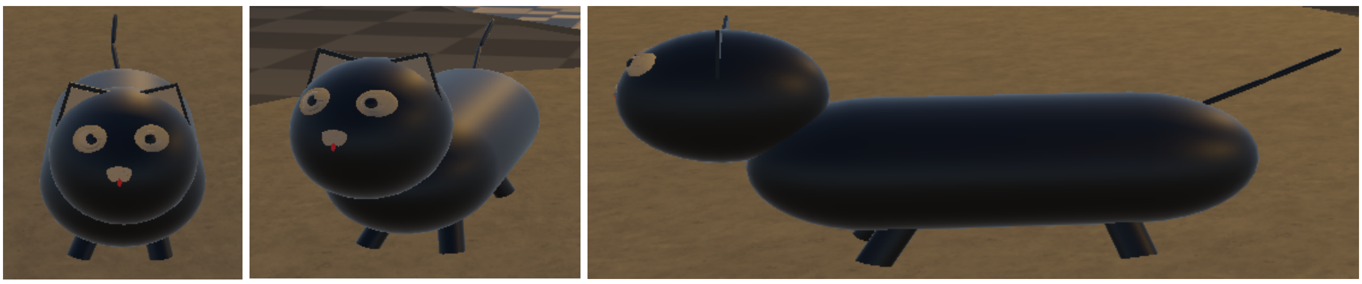

Figure 1: The cat model shown from three different views

2.2. Backstory - Human

The human, shown in **Figure 2** is the cat's owner and lives an ordinary daily life. Compared to the cat, the human perceives the environment in a more practical and structured way, focusing on tasks, tools, and functionality rather than curiosity or play.

2.2.1. Relation to gameplay

The human's gameplay revolves around interacting directly with the environment. While the human cannot access small or elevated spaces as easily as the cat, they can manipulate larger objects, operate machinery, and create paths for the cat. The contrast between the human's practical perspective and the cat's instinctive perspective forms the foundation of the cooperative gameplay.

2.2.2. Controls

The human is controlled with slower and heavier movement compared to the cat. Gameplay focuses on object interaction, environmental manipulation, and positioning rather than agility.

2.2.3. Abilities

- Push and pull heavy objects
- Operate buttons, switches, and machinery
- Create pathways for the cat
- Interact with large environmental objects
- Greater reach and physical strength

Figure 2: The human model shown from three different views

3. Gameplay Design

The following section discusses the design choices made pertaining to gameplay design.

3.1. Progression

The game progresses through a series of increasingly complex cooperative levels, where players gradually learn to combine the abilities of the human and the cat. The initial level inside the home introduce core mechanics such as climbing as the cat and object interaction for the human. Later levels expand on these mechanics with more difficult jumping puzzles and perspective-based challenges. New mechanics are introduced gradually to avoid overwhelming players while still maintaining a sense of progression throughout the game.

3.2. Player retention

Player retention is supported through replayability, cooperation, and mastery. Since each player experiences the world differently, levels encourage replaying with swapped roles to experience the other perspective. Time-based trophies provide additional goals for players who want to improve their performance. The cooperative nature of the game also encourages social engagement, as communication and teamwork are central to the experience.

3.3. Multiplayer

The game is designed as a two-player cooperative experience focused on communication and teamwork. Players must often exchange information because each character perceives and interacts with the world differently. Multiplayer gameplay is designed to encourage discussion rather than individual play, making cooperation necessary for solving puzzles and progressing through levels. The game supports online cooperative play, allowing players to connect remotely and communicate through external voice chat platforms or in-game “communication” systems.

3.4. Asymmetrical Perspectives

The core gameplay mechanic is built around asymmetrical perspectives, where the human and the cat experience the same environment differently. These differences are inspired by how humans and animals naturally perceive the world. Important objects may appear differently depending on the character, hazards may only be noticeable to one player, and certain routes or clues may only be visible from a specific perspective. This system encourages players to communicate clearly and combine their knowledge in order to solve puzzles and navigate the environment together.

4. Game World

This section describes each part of the game world in **Parallax**

4.1. The Playable Lobby

When the players are first dropped into the world, they immediately spawn inside a tall cylinder, with level selection platforms surrounding them. In the middle of the cylinder, there is an elevator, that can transport them between different floors of the cylinder, to access additional levels.

The design for the playable lobby was heavily inspired by the level selection scene in Crash Bandicoot 2.

4.2. The Kitchen

Intended to be the first level of the game is a kitchen/living area in an apartment building. Different everyday items are strewn about, such as a sofa, kitchen table, fridge and a small computer space. Lamps hang from the ceiling and is traversable by the cat, while an electrical locker is situated on the wall, to be interacted with by the human. The level is small in size, and is meant to introduce the players to the different perception of light, between the two characters, the nimbleness of the cat and the most basic of object interaction for both roles.

4.3. The Playground

The playground is the intended second level of the game. The level builds upon the mechanics introduced in the kitchen level while also introduces movable objects. The environment consists of playground equipment such as slides, climbing structures, tunnels, and elevated platforms that both players must navigate together. The Cat can traverse narrow pipes, climb playground structures, and access elevated areas, while the Human interacts with moveable box objects and mechanisms to create traversal opportunities for the Cat. The level is larger and more open than the kitchen, encouraging exploration and communication while still presenting a simple puzzle structure intended for early-game progression.

4.4. The Construction Site

The construction site is the intended third level of the game. It introduces the players to larger levels as well as box mechanics. Here the human has to move boxes around, for the cat to jump on, as well as interact with buttons to move steel beams around. Finally, once the cat has navigated through the two buildings, they must pick up a key for the human. But, for the cat, these all look like toys! The human must communicate to the cat, which is the key they have to pick up.

4.5. The Tower

The Tower is meant to be the fourth level of the game. The level focuses on a combination of many of the gameplay mechanics introduced throughout the earlier levels into larger multi-step cooperative challenges. The environment consists of a tall industrial tower structure with elevators, suspended platforms and killzones for the players to traverse. The Human player is responsible for activating machinery, moving objects, and operating elevators or mechanical systems, while the Cat traverses climbing routes, and small openings inaccessible to the Human. Several sections of the level temporarily separate the players, forcing them to rely on communication in order to solve environmental puzzles and progress further up the tower. The Tower acts as one of the more mechanically demanding levels in Parallax, combining traversal, puzzle solving and asymmetric cooperation.

5. Gestalt

The following section describes the overall gameplay experience the game seeks to deliver.

5.1. End-to-end experience

Parallax is designed as a cooperative experience centered around communication and shared problem-solving. Throughout the game, players must exchange information and rely on one another to navigate environments and overcome obstacles.

The pacing alternates between exploration, puzzle solving, traversal, and moments of tension. Early levels introduce mechanics gradually, while later levels require stronger coordination and communication between players.

The overall experience aims to feel approachable and playful while still rewarding teamwork and cooperation.

5.1.1. First impressions

The first impression of Parallax should immediately communicate the asymmetrical nature of the game. From the beginning, players experience the same environment differently depending on whether they play as the human or the cat.

Human Perspective	Cat Perspective
Slower and practical interaction	Fast and agile traversal
Manipulates larger objects	Accesses hidden pathways
High camera perspective	Lower perspective
Focus on tools and machinery	Focus on exploration and movement

The tutorial sections are designed to introduce mechanics naturally through interaction rather than lengthy instructions.

The game's overall tone is intended to feel curious, playful, and exploration-driven.

5.1.2. Emotions and moods

The game aims to create different emotions throughout the experience.

Gameplay Situation	Intended Emotion
Exploring new areas	Curiosity
Solving puzzles together	Satisfaction
Dangerous traversal	Tension
Helping the other player	Trust
Discovering perspective differences	Wonder

The contrast between the energetic cat perspective and the grounded human perspective reinforces the game's focus on cooperation and communication.

6. Gameplay Mechanics

The following section describes the core gameplay mechanics used throughout Parallax.

6.1. Jumping

Jumping is one of the primary traversal mechanics in the game and is especially important for the cat character. The cat can jump higher, move faster, and access elevated areas that are unreachable for the human.

The human can still jump, but movement is intentionally slower and heavier to reinforce the contrast between the two characters.

6.2. Lifting

Lifting allows the human to interact with heavier environmental objects such as boxes. These objects are often used to create pathways for the cat or solve cooperative puzzles.

The mechanic focuses on environmental interaction rather than precision or speed.

6.3. Pushing

Pushing is used to reposition larger objects within the environment. Players use pushing mechanics to clear pathways, activate pressure plates, or align objects needed for traversal.

Some puzzles require both players to coordinate movement and positioning to progress.

6.4. CatVision

CatVision changes how the environment appears to the cat. Important objects may appear exaggerated, more colorful, or toy-like, reflecting the cat's instinctive interpretation of the world.

This mechanic reinforces the asymmetrical gameplay and encourages players to communicate about what they can see.

6.5. Darkvision

Darkvision allows the cat to navigate dark environments more effectively than the human. Certain areas are intentionally designed with low visibility, requiring the cat player to guide the human through hazards and obstacles.

The mechanic strengthens communication-focused gameplay and creates situations where players must rely on each other to progress.

7. Enemies

Unlike traditional games, Parallax does not focus on combat or direct enemy encounters. Instead, the primary challenges come from environmental hazards, platforming failures, and the pressure created by the level timer.

7.1. Adversaries

The main adversaries in Parallax are the environments themselves. Players are challenged by:

- Failure zones such as falls or dangerous surfaces
- Traversal mistakes during platforming sections
- Time pressure from level completion timers

Rather than fighting enemies directly, players must overcome the level itself through communication, timing, and coordination.

7.2. How to overcome enemies

Players overcome challenges by cooperating and sharing information. Since both characters experience the world differently, communication becomes the most important tool for progression.

Success depends on:

- Coordinating movement and timing
- Warning each other about hazards
- Using each character's abilities effectively
- Improving level knowledge through replayability

The level timer further encourages players to optimize their cooperation and improve their performance over repeated playthroughs.

8. Bonus

The following section outlines the aspects of Parallax that makes the game feel like *Parallax* and not any other game.

8.1. Asymmetrical perspectives

The asymmetrical perspective system is the defining feature of Parallax. Each player experiences the world differently depending on whether they play as the human or the cat.

Player Differences	**Gameplay Purpose**
Different visual information	Encourages communication
Different traversal opportunities	Creates cooperation
Different interactable objects	Makes roles feel unique
Different environmental interpretations	Improves replayability

Swapping player roles provides a noticeably different gameplay experience and encourages replaying levels from the opposite perspective.

8.2. Replayability

Replayability is supported through both progression systems and cooperative mastery.

Replay Element	**Purpose**
Swapping character roles	Experience the other perspective
Time-based trophies	Encourage optimization
Developer challenge trophies	Provide optional difficulty

The goal is for players to revisit earlier levels with improved teamwork and knowledge of the environment.

9. Money

The project is developed as a student bachelor project with a limited budget.

The majority of development tools are either:

- Already available through the university
- Free for educational use
- Open-source
- Covered by existing student licenses

Expense	**Estimated Cost**
Unity assets and plugins	Free
Sound effects and music	Royalty free
Server hosting for multiplayer testing	Free
Version control and collaboration tools	Free / Student
Art software	Already owned

The project scope is intentionally limited to remain realistic within the timeframe of a bachelor project.

The development prioritizes gameplay mechanics and cooperative systems over large content quantity.

9.1. Monetization

The game has no monetization planned, as it is intended to be a prototype only.

10. Deviations From The Plan

During development, several aspects of the original project plan changed as the scope and technical complexity of the game became clearer.

One of the largest deviations concerned the multiplayer architecture. The original plan involved a more server-authoritative networking approach, but this was simplified during development to better fit the project scope and timeframe. Instead, the project focused on creating a stable cooperative experience while documenting the limitations and lessons learned from the chosen implementation.

The original project proposal also described more advanced asymmetrical perception systems, where both players would experience significantly different versions of the same environment. While parts of this concept were implemented successfully, some mechanics were simplified to maintain gameplay clarity and reduce development complexity.

Additionally, several planned levels and environmental interactions were either reduced in scope or redesigned during production. This was done to prioritize polish, stability, and the core cooperative gameplay loop over a larger quantity of content.

Despite these changes, the project maintained its original focus on asymmetrical cooperative gameplay centered around communication and teamwork.

11. Challenges to Overcome

Throughout development, the project presented several technical and design-related challenges that influenced both the scope and final implementation of the game.

One of the main challenges, were the interaction system. Since gameplay depended heavily on cooperation and the ability to move stuff as the human, the goal of the interaction system was to make it feel “smooth”. physics-based interactions and shared object states, made it difficult to fine-tune the movement while dragging large boxes.

Another major challenge involved designing asymmetrical gameplay that remained intuitive for both players. Since the human and cat experience the world differently, puzzles needed to provide meaningful information to both players without causing confusion or making one role feel less important than the other.

Level design also became more challenging over time as puzzle complexity increased. Many ideas that worked conceptually were difficult to communicate clearly during gameplay, especially when both players had incomplete information. This required extensive playtesting and simplification of several mechanics and puzzle layouts.

Finally, scope management became an important part of development. As the project progressed, the group had to continuously prioritize core gameplay systems, stable multiplayer functionality, and polished level design over additional mechanics or content.

Despite these challenges, the project successfully achieved its primary goal of creating a cooperative experience focused on communication, asymmetrical perspectives, and teamwork.

12. Reflection

The development of Parallax provided practical experience with both technical implementation and collaborative software development. Throughout the project, the group worked with multiplayer networking, systems programming, level design, 3D development, and user-focused design principles within a larger Unity project.

One of the most important lessons learned during development was the complexity involved in creating asymmetrical cooperative gameplay. Designing mechanics where both players experience the same environment differently required careful balancing to ensure that communication remained meaningful without becoming frustrating or unclear.

The project also highlighted the importance of iterative development and playtesting. Several mechanics and puzzle ideas that initially appeared strong in theory required redesigns after testing revealed issues with pacing, readability, or player understanding. This emphasized the value of testing gameplay systems early and continuously throughout development.

From a technical perspective, the project provided insight into the challenges of multiplayer synchronization and game architecture. Networked movement, interactable objects, and cooperative puzzle systems introduced technical constraints that influenced both design decisions and implementation strategies.

Additionally, the project strengthened the group's experience with teamwork, task delegation, version control, and project planning. Coordinating development across multiple systems and disciplines required continuous communication and prioritization throughout the project period.

Overall, the project demonstrated how software engineering principles can be applied within game development while also showing how games can be designed to encourage cooperation, communication, and shared problem-solving between players.